

Inequality and Misallocation under Production Networks

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Research Question

How do changes in the distributions of **factor income** and **consumption expenditure** affect aggregate **TFP**?

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Independent of
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Traditional route: demand composition affects Y through nominal/financial frictions. Redistribution from high $\rightarrow$ low MPC households shifts AD ; output falls when:

1. **Downward wage rigidity** $\Rightarrow$ Keynesian unemployment, $Y \downarrow$ (*Auclert & Rognlie, 2018*).
2. **Zero Lower Bound (ZLB)** $\Rightarrow$ real rate too high, capital market can't clear $\Rightarrow Y \downarrow$ (*Mian, Straub & Sufi, 2020*).

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This project: AD recomposition $\Rightarrow Y$ via **TFP**

In economies with:

1. Dual heterogeneity
2. Rebated distortions

Motivation & Literature

Upper Decile have a higher share in
Education and Entertainment
vs.

The Rest have a higher share in
Shelter, Utilities, Healthcare

Data from Consumer Expenditure Survey

Pricing Inequality

Sangani (2024): Using transactions matched to wholesale costs, higher-income households pay higher retail markups (within-store and across-product components).

Consumption Inequality

Aguiar & Hurst (2007): price dispersion

Aguiar & Bils (2015): $\uparrow$ consumption inequality

Increasing income and
consumption inequality

Approach

Contribution

Δ Consumption Distribution & Δ Demand Structure



Δ Income Distribution

Theorem 1

Approach

Contribution

Δ Consumption Distribution & Δ Demand Structure



Δ Income Distribution

Δ Factor Income Shares

Theorem 1

Approach

Δ **Consumption
Distribution** & Δ **Demand
Structure**



Δ **Income Distribution**

Contribution

Bigio & La'O (2020)

- (i) Rep Household
- (ii) Efficient Equilibrium

$\downarrow$ Δ **Factor Income Shares**

Theorem 1

Approach

Δ **Consumption Distribution** & Δ **Demand Structure**



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My Model

- (i) Het Households
- (ii) Any Equilibrium

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Δ **Income Distribution**



Δ **Misallocation**



Δ **TFP**

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Theorem 1

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- (i) Rep Household
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- (i) Het Households
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Theorem 1

Baqaei & Farhi (2020)

Rep Household



My Model

Het Households

Theorem 2

Theorem 3

Centralizing households' decisions in a **constrained social planner** yields a different Δ TFP in which distributional variations are first-order *neutral*.

Theorem 4 - Incidence

Decomposition for **household-specific real consumption** that measures heterogeneous exposure to growth and network reallocation (incidence) across households.

Empirical Results

- **Distributions explain $\approx 20\%$ of U.S. TFP fluctuations (1997–2021).**
- Budget-neutral lump-sum taxes/transfers can raise TFP when targeted toward **high expenditure centrality** households, mitigating network bottlenecks. These gains can be achieved while improving household-level real consumption for low- and middle-income households.

Literature Review

- **Digggregated National Accounts**

Cantillon (1756), Quesnay (1758), Leontief (1928), Meade & Stone (1941), Kuznetz (1946), Stone (1961), Andersen et al. (2022)

- **Production Networks**

Hulten (1978), Long & Plosser (1983), Gabaix (2011), Jones (2011, 2013), Acemoglu et al. (2012), Baqaee (2018), Baqaee & Farhi (2019, 2020, 2023), Bigio & La'O (2020)

- **Growth Accounting**

Solow (1957), Domar (1961), Jorgenson et al. (1987), Basu & Fernandez (2022), Petrin & Levinsohn (2012), Baqaee & Farhi (2020)

- **2-sided heterogeneity economies - Similar environments, different statistics**

Davila & Schaab (2025), Baqaee & Burnstein (2025)

Environment

- Non-parametric CRS economy in production and consumption

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- **Production:** Firm $i \in \mathcal{N}$ produces using

$$y_i = A_i Q_i(L_i, X_i) \quad \begin{cases} L_i : \text{Primary factor bundle} \\ X_i : \text{Intermediate input bundle} \end{cases}$$

$$\begin{aligned} L_i &= Q_i^\ell(\{\ell_{ih}\}_{h \in \mathcal{H}}) \\ X_i &= Q_i^\ell(\{\mathbf{x}_{ij}\}_{j \in \mathcal{N}}) \end{aligned} \quad \begin{cases} \ell_{ih} : \text{Factor supplied by household } h \\ \mathbf{x}_{ij} : \text{Intermediate input supplied by firm } j \end{cases}$$

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Firm i solves

$$\underset{y_i, \{\ell_{ih}\}_{h \in \mathcal{H}}, \{x_{ij}\}_{j \in \mathcal{N}}}{\text{Min}} \quad \text{Cost}_i = \sum_{h \in \mathcal{H}} w_h \ell_{ih} + \sum_{j \in \mathcal{N}} p_j x_{ij} \quad \text{subject to}$$

$$1 = A_i Q_i(L_i, X_i) \quad \text{and} \quad mc_i = \mu_i p_i \quad \rightarrow \quad \text{Cost}_i = \mu_i p_i y_i$$

where μ_i stands for the **inverse markup** with $0 < \mu_i \leq 1$

Environment

- **Consumption:** Household $h \in \mathcal{H}$

$$\underset{\{C_h, \{C_{hi}\}_{i \in \mathcal{N}}, L_h\}}{\text{Max}} \quad U_h(C_h, L_h) \quad \text{subject to}$$

$$C_h = Q_h^c(\{C_{hi}\}_{i \in \mathcal{N}}) \quad \text{and} \quad \underbrace{p_h^c C_h = \sum_{i \in \mathcal{N}} p_i C_{hi}}_{\text{Expenditure}} \leq \underbrace{w_h L_h}_{\text{Factor Income}} + \underbrace{\sum_{i \in \mathcal{N}} \kappa_{ih} \pi_i}_{\text{Corporate Income}} + \underbrace{T_h}_{\text{Lump Sum Transfers}}$$

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- **Market Clearing:**

$$y_i = \sum_{h \in \mathcal{H}} C_{hi} + \sum_{j \in \mathcal{N}} x_{ji} \quad L_h = \sum_{i \in \mathcal{N}} \ell_{ih} \quad \sum_{h \in \mathcal{H}} T_h = 0$$

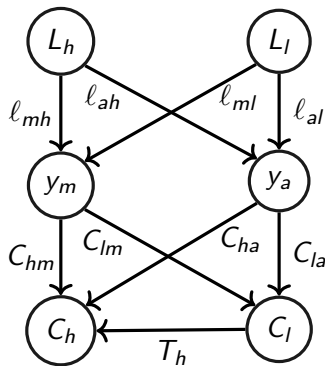
Goods Market

Factor Market

Balanced Budget

2-firm & 2-household economy

Two firm manufacturing (m) and agriculture (a)



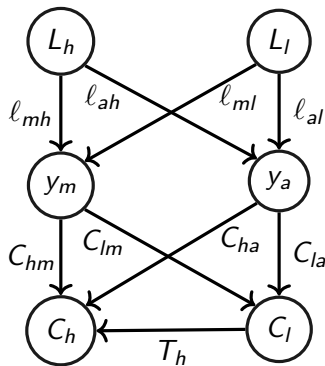
$$y_i = \sqrt{\ell_{ih} \ell_{il}} \quad \text{for} \quad i \in \{m, l\}$$

$$C_b = C_{bm}^{\beta_b} C_{ba}^{1-\beta_b} \quad \text{for} \quad b \in \{h, l\}$$

$$\beta_h > \beta_l \quad \mu_a > \mu_m \quad \kappa_{ib} = \frac{1}{2}$$

2-firm & 2-household economy

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Shock: Starting from $T_h = 0$, we are going to understand how $d\tau_h$ has an effect on this economy, where $\tau_h = T_h / \text{GDP}$.

Four Theorems

Theorem 1

How the **factor income distribution** changes in these economies?

Theorem 2

How aggregate **TFP** (efficiency) responds to variations in the distributions of **factor income** and **consumption expenditure**?

Theorem 3

How these results differ relative to a **constrained social planner** ($\approx$ representative household) economy?

Theorem 4

How each household's **real consumption** responds idiosyncratically to shocks?

Factor Income Distribution

- Factor income

$$\underbrace{\Lambda_h}_{\text{Factor Share}} = \frac{w_h L_h}{GDP} = \sum_{b \in \mathcal{H}} \underbrace{\mathcal{C}_{bh}}_{\text{Household to Factor Centralities}} \underbrace{\chi_b}_{\text{Expenditure Share Endogenous}}$$

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- Where

$$\underbrace{\mathcal{C}}_{\text{Household to Factor Centralities}} = \underbrace{\beta}_{\text{Matrix of Consumption Expenditure}} \underbrace{(I - \Omega_x)^{-1}}_{\text{Revenue-Based Leontief Inverse Matrix}} \underbrace{\Omega_\ell}_{\text{Revenue-Based Factor Network}}$$

$$\mathcal{C}_{hb} = \beta_h \frac{\mu_m}{2} + (1 - \beta_h) \frac{\mu_a}{2} \quad \rightarrow \quad \Lambda_h = \Lambda_l = \frac{\mu_a}{2} + \frac{1}{4} (\beta_h + \beta_l) (\mu_m - \mu_a)$$

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Cost-based objects for **downstream propagation of costs**, e.g.,

$$\text{Value Added Shares} \quad \tilde{\Lambda}_h = \tilde{\Lambda}_l = \frac{1}{2}$$

Theorem 1

$$\Lambda_h = \mathcal{C}_{hh} \chi_h + \mathcal{C}_{lh} \chi_l$$

$$\mathcal{C}_{bh} = \beta_b \psi_{mh}^l + (1 - \beta_b) \psi_{ah}^l$$

$$\psi_{ih}^l = \frac{\mu_i}{2}$$

Theorem 1

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$$d \Lambda_h = \underbrace{\mathcal{C}_{hh} d \chi_h + \mathcal{C}_{lh} d \chi_l}_{\text{Distributive Income}_h} + \underbrace{\chi_h d \mathcal{C}_{hh} + \chi_l d \mathcal{C}_{lh}}_{\text{Income Centrality}_h}$$

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$$\begin{aligned} \text{Income Centrality}_h &= \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \sum_{b \in \mathcal{H}} \chi_b d\beta_{bi}}_{\text{Final Demand Recomposition}_h} + \underbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i d\tilde{\Omega}_{ih}^\ell}_{\text{Factor Demand Recomposition}_h} \\ &+ \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \sum_{j \in \mathcal{N}} \mu_j \lambda_j d\tilde{\Omega}_{ji}^x}_{\text{Intermediate Demand Recomposition}_h} + \underbrace{\sum_{i \in \mathcal{N}} \psi_{ih}^\ell \lambda_i d \log \mu_i}_{\text{Competitive Income}_h} \end{aligned}$$

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Bigio & La'O (2020)

For $\Gamma = \sum_{h \in \mathcal{H}} \Lambda_h$

$$\frac{\partial \log \Gamma}{\partial \log \mu_i} \approx \lambda_i \quad \text{with} \quad \lambda_i = \frac{p_i y_i}{GDP}$$

around first-best

Theorem 2

$$Y = Q_Y \left(\{C_h\}_{h \in \mathcal{H}} \right) = TFP F \left(\{L_h\}_{h \in \mathcal{H}} \right)$$

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$$d \log TFP \approx \overbrace{\sum_{i \in \mathcal{N}} \tilde{\lambda}_i}^{\text{Technology}} d \log A_i$$

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Competitiveness
Distributional Reallocation

$$\text{Distributional Reallocation} = - \sum_{h \in \mathcal{H}} \underbrace{\left(\tilde{\Lambda}_h / \Lambda_h \right)}_{\delta_h} d \Lambda_h$$

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Distributional Reallocation

Distortion Centralities

$$\delta_h : \frac{\text{Value Added}_h}{\text{Factor Income}_h}$$

Measures how **undervalued** a factor is

$$\text{Distributional Reallocation} = - \sum_{h \in \mathcal{H}} \underbrace{\left(\tilde{\Lambda}_h / \Lambda_h \right)}_{\delta_h} d \Lambda_h$$

Theorem 2

**Distributional
Reallocation** $= - \sum_{h \in \mathcal{H}} \delta_h d \Lambda_h$

Introduce $d \Lambda_h$ from Theorem 1

Theorem 2

$$\text{Distributional Reallocation} = - \sum_{h \in \mathcal{H}} \delta_h d \Lambda_h$$

$$= - \overbrace{\sum_{h \in \mathcal{H}} M_h d \chi_h}^{\text{Distributive TT}}$$

- **Expenditure Centralities**: $M_h = \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} \delta_b \rightarrow \text{TFP} \uparrow \text{ as expenditure shifts from high to low } M$

Theorem 2

Distributional Reallocation $= - \sum_{h \in \mathcal{H}} \delta_h d \Lambda_h$

$$= - \overbrace{\sum_{h \in \mathcal{H}} M_h d \chi_h}^{\text{Distributive TT}} - \overbrace{\sum_{j \in \mathcal{N}} F_i \left(\sum_{h \in \mathcal{H}} \chi_h d \beta_{hi} + \sum_{j \in \mathcal{N}} \mu_j \lambda_j d \tilde{\Omega}_{ji}^x \right)}^{\text{Final and Intermediate Demand TT}}$$

- **Expenditure Centralities**: $M_h = \sum_{b \in \mathcal{H}} \mathcal{C}_{hb} \delta_b \rightarrow \text{TFP} \uparrow$ as expenditure shifts from high to low M
- **Revenue Centralities**: $F_i = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell \delta_h \rightarrow \text{TFP} \uparrow$ as costs shift from high to low F

Theorem 2

Distributional Reallocation

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- **Distortion Centralities**: $\delta_h = \frac{\tilde{\Lambda}_h}{\Lambda_h} \rightarrow \text{TFP} \uparrow$ as factor costs shift from high to low δ

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Back to our Simple Economy

- Both $\tilde{\Lambda}_h = \tilde{\Lambda}_l$ and $\Lambda_h = \Lambda_l$.

Hence **distortion centralities** satisfy

$$\delta_h = \delta_l = \frac{1}{2\Lambda_h}$$

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$$F_a - F_m = \frac{\mu_a - \mu_m}{2\Lambda_h} > 0$$

Expenditure shifting towards manufacturing raises TFP

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expenditure shifting towards h households raises TFP

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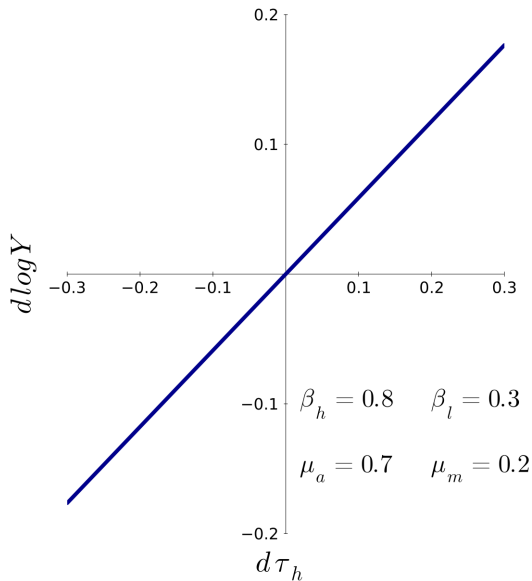
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expenditure shifting towards h households raises TFP

- In response to the $d\tau_h$ shock we have that the only endogenous variation is given by

$$d\chi_h = d\tau_h$$



$$d \log Y = - (M_h - M_l) d \chi_h$$

$$= \frac{(\beta_h - \beta_l)(\mu_a - \mu_m)}{\mu_a + \frac{1}{2}(\beta_h + \beta_l)(\mu_m - \mu_a)} d \tau_h$$

- Reallocation of expenditure through lump-sum taxes from l to h households shifts aggregate expenditure from a to m
- Relative labor demand from m increases, moving workers from a to m
- Workers have a higher marginal productivity in m , which increases TFP

Theorem 3

Constraint Social Planner

$$\text{Max}_{\{C_h, L_h, \{C_{hi}\}_{i \in \mathcal{N}}, T_h\}_{h \in \mathcal{H}}} W(Y, L) \quad \text{s.t.}$$

$$Y = Q_Y(\{C_h\}_{h \in \mathcal{H}}), \quad L = F(\{L_h\}_{h \in \mathcal{H}})$$

$$p_Y Y \leq \sum_{h \in \mathcal{H}} w_h L_h + \sum_{i \in \mathcal{N}} \pi_i + T_h$$

Theorem 3

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Factor supply is chosen to satisfy

$$\delta_h = \frac{\tilde{\Lambda}_h}{\Lambda_h} = \Gamma^{-1} \quad \forall h \in \mathcal{H}$$

- The margins of variation on aggregate output coming from the reallocation of income across factors and expenditure across households are shut off

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$$\delta_h = \frac{\tilde{\Lambda}_h}{\Lambda_h} = \Gamma^{-1} \quad \forall h \in \mathcal{H}$$

Consumption is chosen to satisfy

$$\frac{p_i}{p_Y} = \frac{\partial Y}{\partial C_h} \frac{\partial C_h}{\partial C_{hi}} \quad \begin{matrix} \forall h \in \mathcal{H} \\ \forall i \in \mathcal{N} \end{matrix}$$

- The margins of variation on aggregate output coming from the reallocation of income across factors and expenditure across households are shut off

Theorem 3

The only channels that are still open are those that depend on firm-level allocation

$$d \log TFP = \sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log A_i + \sum_{i \in \mathcal{N}} \tilde{\lambda}_i d \log \mu_i - d \log \Gamma$$

where $\Gamma = \sum_{h \in \mathcal{H}} \Lambda_h$ stands for the aggregate factor share, and

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$$d \Gamma = \sum_{i \in \mathcal{N}} \psi_i^\ell \lambda_i d \log \mu_i + \sum_{i \in \mathcal{N}} \mu_i \lambda_i d \omega_i^\ell + \sum_{i \in \mathcal{N}} \psi_i^\ell \left(d \beta_i + \sum_{j \in \mathcal{N}} \mu_j \lambda_j d \tilde{\Omega}_{ji}^x \right).$$

where $\psi_i^\ell = \sum_{h \in \mathcal{H}} \psi_{ih}^\ell$

Positional Terms of Trade_{*h*}

Theorem 4

$$C_h = Q_h^c \left(\{C_{hi}\}_{i \in \mathcal{N}} \right) = PTT_h f_h \left(\{L_b\}_{b \in \mathcal{H}} \right),$$

Positional Terms of Trade_h

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$$C_h = Q_h^c \left(\{C_{hi}\}_{i \in \mathcal{N}} \right) = PTT_h f_h \left(\{L_b\}_{b \in \mathcal{H}} \right),$$

$$d \log TFP = \sum \chi_h d \log PTT_h$$

Decomposition of TFP
across households

Positional Terms of Trade_h

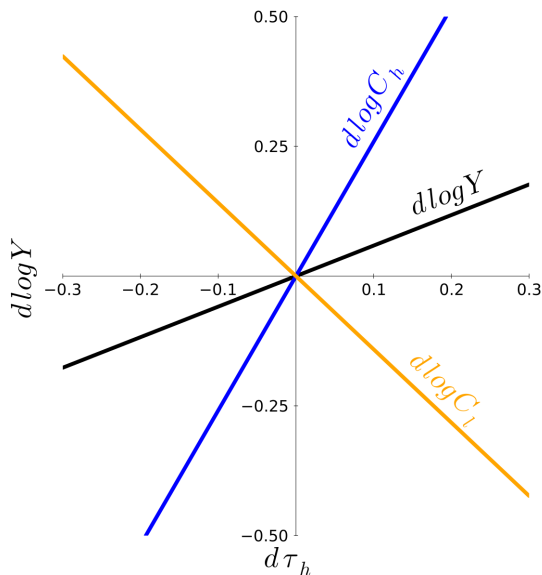
Theorem 4

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$$d \log TFP = \sum \chi_h d \log PTT_h$$

Decomposition of TFP
across households

$$\begin{aligned} d \log PTT_h &\approx \overbrace{\sum_{i \in \mathcal{N}} \widetilde{\mathcal{B}}_{hi} d \log A_i}^{\text{Technology}_h} + \overbrace{\sum_{i \in \mathcal{N}} \widetilde{\mathcal{B}}_{hi} d \log \mu_i}^{\text{Competitiveness}_h} + d \log \chi_h \\ &- \overbrace{\sum_{b \in \mathcal{H}} M_{b|h} d \chi_b}^{\text{Distributive } TT_h} - \overbrace{\sum_{b \in \mathcal{H}} \chi_b \sum_{i \in \mathcal{N}} F_{i|h} d \beta_{bi}}^{\text{Final Demand } TT_h} - \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{j \in \mathcal{N}} F_{j|h} d \tilde{\Omega}_{ij}^x}^{\text{Intermediate Demand } TT_h} \\ &- \overbrace{\sum_{i \in \mathcal{N}} \mu_i \lambda_i \sum_{b \in \mathcal{H}} \delta_{b|h} d \tilde{\Omega}_{ib}^\ell}^{\text{Labor Demand } TT_h} - \overbrace{\sum_{i \in \mathcal{N}} \lambda_i F_{i|h} d \log \mu_i}^{\text{Competitive } TT_h} \end{aligned}$$



- Reallocation of workers from a to m favors h and hurts l
- The increase in C_h more than dominates the loss in C_l
- Real consumption effects from lump-sum transfers are not zero-sum

Data Sources I - Representative Household

Industry Level Production Accounts

- Sector-level *TFP* by year ($\approx A_i$)
- Sector-level **capital compensation** across 9 forms of capital:
 - (1) commercial equipment, (2) computer equipment,
 - (3) R&D capital services, (4) software capital services,
 - (5) artistic originals, (6) institutional equipment, (7) structures, (8) transportation, and (9) other

Input Output Tables

- Labor Costs by sector
- Intermediate Input Costs by sector
- Final consumption by sector
- Using sectoral **capital** + labor + intermediate costs and sales, I estimate $\mu_{i,t}$

Model with 66 sectors with intermediate inputs + Household has 10 different sources of income (Labor + 9 types of capital + Profits)

Model Comparison - Against BLS's TFP estimate

Metric	Labor No Networks	Labor Networks	Lab & Cap No Networks	Lab & Cap Networks
Mean Absolute Error	$6.2e^{-3}$	$1.5e^{-2}$	$6.2e^{-3}$	$4.7e^{-3}$
Root Mean Squared Error	$7.4e^{-3}$	$1.9e^{-2}$	$7.3e^{-3}$	$5.8e^{-3}$
Bias (Mean Error)	$5.4e^{-3}$	$-8.9e^{-3}$	$5.3e^{-3}$	$2.1e^{-4}$
Correlation	0.871	0.702	0.873	0.875
Nash-Sutcliffe Efficiency	0.417	-2.99	0.425	0.627
Index of Agreement	0.815	0.646	0.819	0.923
Theil's U Statistic	0.353	0.459	0.349	0.209
Kling-Gupta Efficiency	.319	-0.679	0.324	0.720

Model with **Labor + Capital + Networks** captures the best match. **Aggregating granular sources of variation in this economy captures quite well aggregate variation. Caveat:** BLS's TFP is a Solow residual (blackbox); instead, here TFP uses previous definition.

Data Sources II – State-Level Model

State-Level PCE

- Map state PCE across NIPA PCE commodities
- Bridge to NAICS classification
- Use IO Make matrix to build industry-level consumption vector
- Capital and firm owners assigned residual consumption to match aggregates

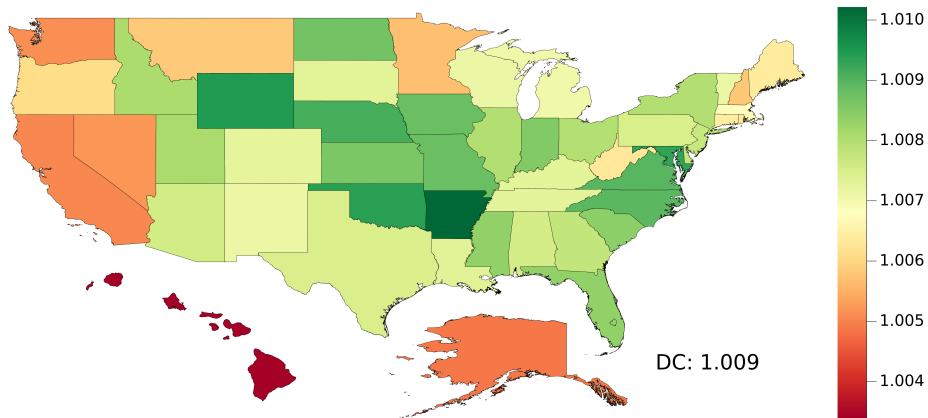
QCEW

- Sector-level labor hiring by state
- Geographic variation in labor income exposure

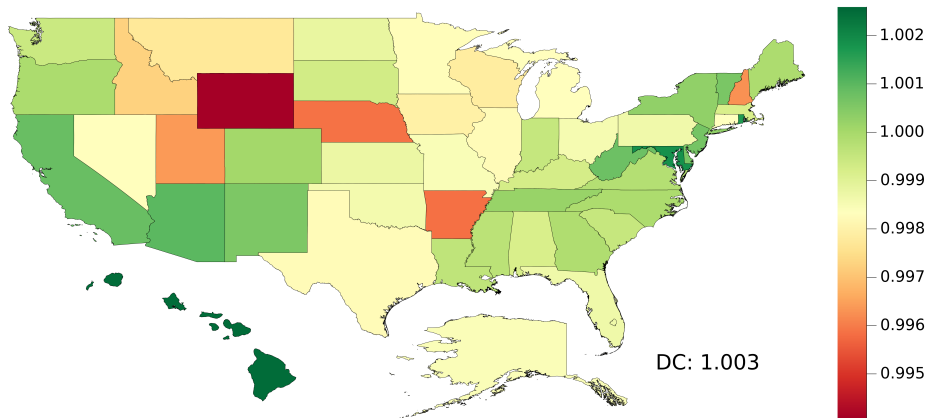
Model: 66 sectors with input–output linkages + 61 households (50 states + DC labor, 9 capital owners, firm owners)

Negligible effect on TFP fit: near-zero movements in performance metrics.

Expenditure Centralities M in 1997 - χ from high-to-low $\uparrow$ TFP



Expenditure Centralities M in 2021 - χ from high-to-low $\uparrow$ TFP



Data Sources III - Occupation-Level Model

Consumer Expenditure Survey

- Map household expenditures (UCC) into NIPA PCE commodities (Markevych, 2025)
- Bridge commodities to NAICS industries
- For each household, use 100 nearest-income neighbors to impute NAICS-level spending
- Apply IO Make matrix to build industry consumption vector
- Generate consumption profiles by income group
- Assign residual spending to capital and firm owners to match aggregates

Occupation and Employment Wage Statistics

- Sector level intensity in labor hiring by occupation
- Occupational variation in labor income exposure by industry

Model with 66 sectors and 843 households (833 occupations + 9 capital owners + firm owners). Labor links from OEWS; consumption based on expenditure vector nearest to average occupational income.

Model Comparison - Against BLS's TFP estimate

Table: Model Comparison: Representative Household vs. Occupation-Level Model

Metric	Rep-Hou Model	Occupation-Level Model
Mean Absolute Error	0.0045	0.0065
Root Mean Squared Error	0.0061	0.0079
Bias (Mean Error)	-0.0002	0.0001
Correlation	0.8598	0.8263
Nash-Sutcliffe Efficiency	0.5962	0.3248
Index of Agreement	0.9141	0.8788
Theil's U Statistic	0.2165	0.2675
Kling-Gupta Efficiency	0.7235	0.5411

Occupation-Level model performs well.

Covariance Decomposition

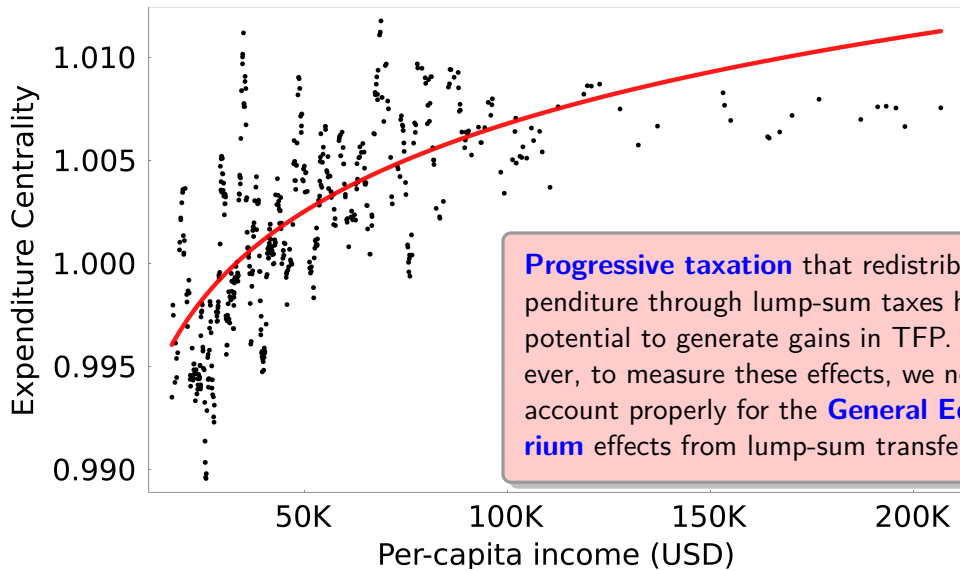
Technology	Competitiveness	Distributional Reallocation	Residual
116.86%	-17.52%	17.48%	-16.82%

(1) Technology is the main source of variation; (2) Competitiveness negative $\Rightarrow$ **procyclical markups**; (3) Distributional Reallocation offsets competitiveness; (4) Last two channels explain about **1/5 of the variation**.

Distributive TT	Final TT	Intermediate TT	Factor TT	Competitive TT
0.56%	-1.17%	23.24%	-23.42%	-16.69%

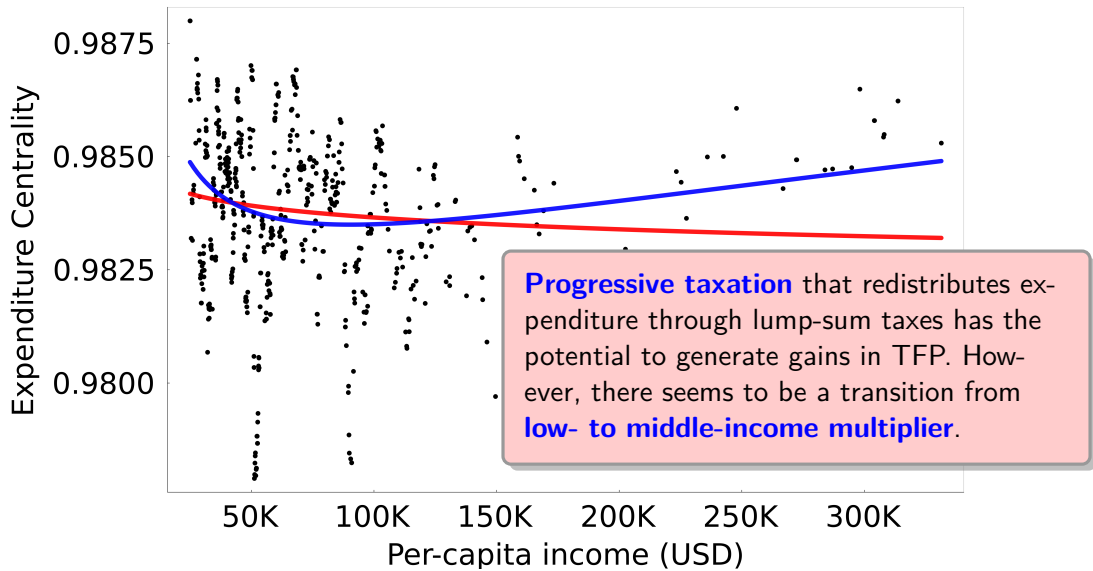
(1) Variation mainly from intermediate, factor, and competitive TT terms; (2) Intermediate input intensity is **procyclical** (Ghassibe, 2022); (3) Consumption expenditure variation is largely irrelevant.

Expenditure Centralities M in 2008 – χ from high-to-low $\uparrow$ TFP



Progressive taxation that redistributes expenditure through lump-sum taxes has the potential to generate gains in TFP. However, to measure these effects, we need to account properly for the **General Equilibrium** effects from lump-sum transfers.

Expenditure Centralities M in 2021 – χ from high-to-low $\uparrow$ TFP



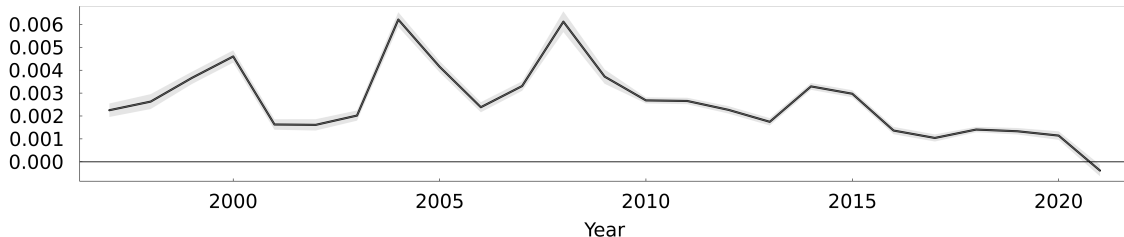
Positive Correlation is a Constant Across Time

In the regression

$\ln M_{h,t} = \alpha_t + \beta_t \ln \text{per worker income}_{o,t} + \epsilon_{o,t}$

the β_t is given by

Lower-income workers have higher expenditure exposure to more distorted supply chains. $\approx$ but not necessarily $\equiv$ to higher markup exposure



Parametric Model with GE effects

Model and Parameterization

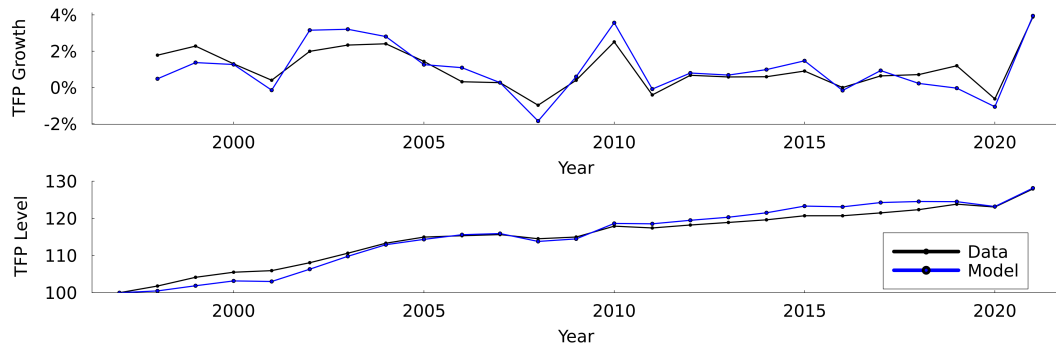
- **Progressive lump-sum taxation can raise TFP** via the **Distributive TT** (*partial equilibrium*).
- To capture **general equilibrium** effects on TFP, estimate a **nested CES** economy with:
 1. θ : elasticity between L and X
 2. θ_ℓ : elasticity within L
 3. θ_x : elasticity within X
 4. ρ : elasticity across C
- **Two-stage GMM** to match $\widehat{TFP}$ treating $\widehat{A}$ and $\widehat{\mu}$ as exogenous. **Stage 1**: global (differential evolution), bounds $[0, 3]$; **Stage 2**: local (L-BFGS, forward diff).

GMM Regression Results

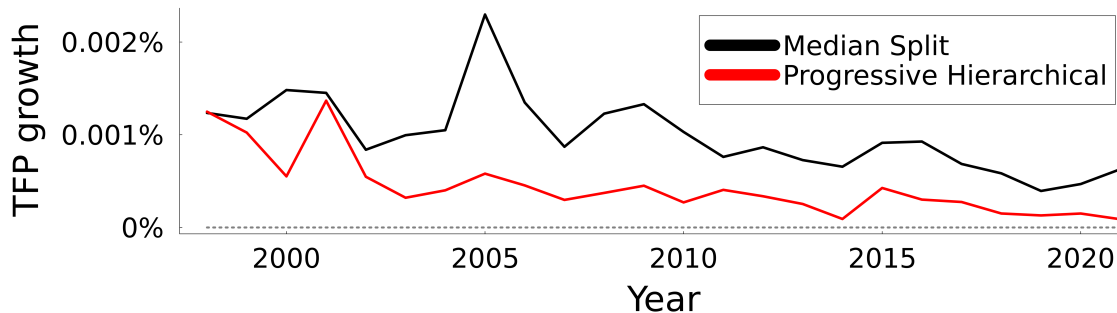
Par	Mean	Sdv	Best	95% CI
θ	0.076	0.058	0.01	[0.005, 0.234]
θ_ℓ	1.482	0.825	0.940	[0.282, 2.855]
θ_x	1.954	0.425	2.398	[1.015, 2.710]
ρ	0.466	0.346	0.073	[0.015, 1.249]

Complementarity *almost everywhere*. Consistent with **Boehm, Flaen & Nayar (2014)** and **Atalay (2017)** who only estimate θ and θ_x .

Correlation: 0.89 & Index of Agreement: 0.93



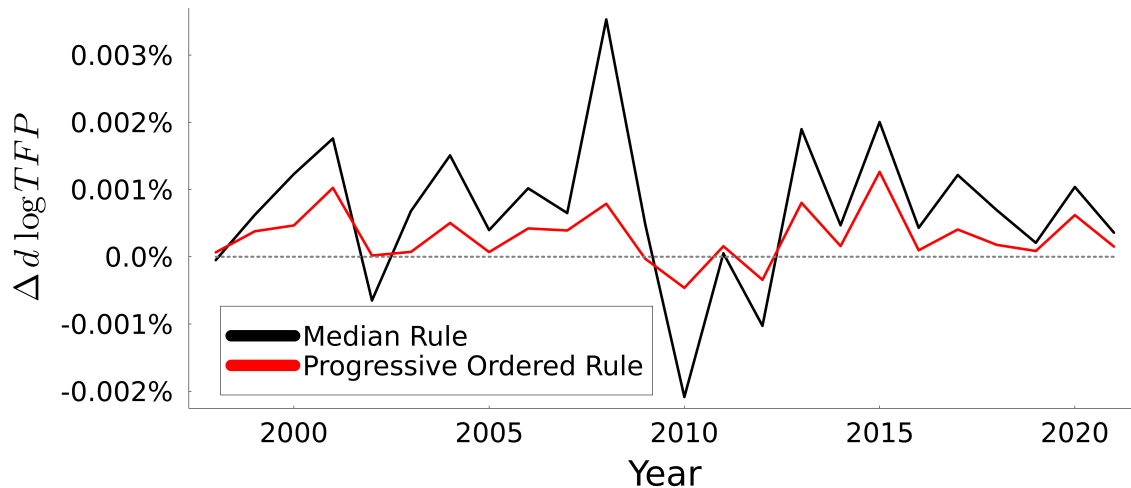
\$1000 lump-sum transfer – 2 reallocation rules



\$1000 as a proportion of 1997 GDP. **Median rule:** Identify median, above median gets taxed, below median receives a transfer.

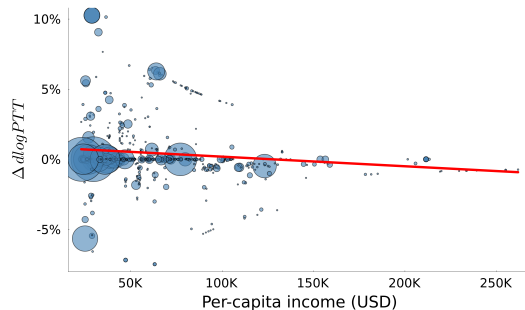
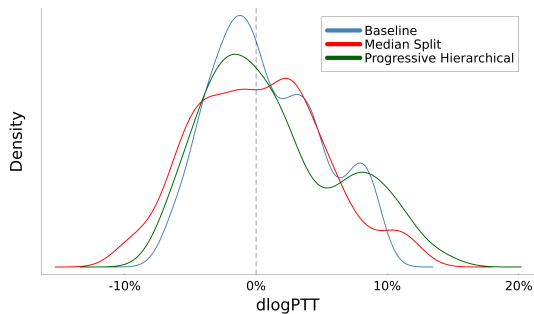
Progressive ordered rule: start with highest M and find lowest M conditional on progressivity.

$d \log TFP$ difference when other shocks are accounted



For most years, both heuristic rules create a positive difference relative to the model-predicted increase in TFP.

$d \log PTT$ Distributions and Progressive Differential by income (2019)



Conclusion

- First comprehensive study with joint heterogeneity in multisector economies with distortions and input-output networks
- **Theoretical Contribution** in production network + distortions + heterogeneous households:
 - Variation of the income distribution
 - Variations for TFP
 - Variations for PTT
- **Empirical Contribution:** First implementation of a production network model with household heterogeneity for the US
 - Lump-sum taxation has the potential to increase TFP by improving the reallocation of resources in response to expenditure and aggregate demand recomposition

Upper Decile vs The Rest

(Consumer Expenditure Survey 2021)

Higher Expenditure Share in

- **Education**: 3.4% vs 1.3%
- **Entertainment**: 6.5% vs 4.9%
- **Pensions**: 17.4% vs 9.1%
- **Lodging**: 2.6% vs 1.1%

Lower Expenditure Share in

- **Shelter**: 17.6% vs 20.5%
- **Home Food**: 5.9% vs 8.5%
- **Utilities**: 4.1% vs 7.0%
- **Healthcare**: 6.2% vs 8.3%

From 2004 to 2019

Income share for top quintile ↑ from 48% to 53%